

Hide menu

Approximation Algorithms and Approximation Ratios: Introduction

Video: Introduction to Approximation Algorithms6 min

Quiz: Approximation Algorithm BasicsSubmitted

Jobshop Minimum Makespan Scheduling

Vertex Cover

Maximum Satisfiability Problem

Problem Set # 3

🎉 Congratulations! You passed!

Grade received 100%

Latest Submission Grade 100%

To pass 80% or higher

Go to next item

Approximation Algorithm Basics

Review Learning Objectives

Submit your assignment

Due Feb 25, 11:59 PM IST

Receive grade

To Pass 80% or higher

1. We have designed an efficient 2-factor approximation algorithm for a minimization problem. Running the approximation algorithm on an instance yields a solution with objective value 12. What can we say about the optimum?

4 / 4 points

☐ The optimal solution can be 1.

☒ The optimal solution is in the range $6 \leq \text{OPT} \leq 12$

👉 Correct

Correct. The optimal solution value must be less than or equal to the approximation algorithm but the factor of $\alpha = 2$ guarantees that its value is at least 6.

☐ The optimal solution is in the range $12 \leq \text{OPT} \leq 24$

☐ None of the other options provided here can be said to hold about the optimal solution without further information about the problem.

2. We have designed an efficient 3-factor approximation algorithm for a **maximization** problem. Running the approximation algorithm on an instance yields a solution with objective value 12. What can we say about the optimum?

1 / 1 point

☐ The optimal solution can have the value 995

☐ The optimal solution is in the range $4 \leq \text{OPT} \leq 12$

☒ The optimal solution is in the range $12 \leq \text{OPT} \leq 36$

👉 Correct

Correct

☒ The optimal solution may be the same as that of the approximation algorithm.

👉 Correct

Correct: this is possible in some cases.

👍 Like

👎 Dislike

📄 Report an issue

Your grade 100%

View Feedback

We keep your highest score